



In this page,

$C^\infty = \{f: \mathbb{R} \rightarrow \mathbb{C} \mid f \text{ is infinitely differentiable}\}$, Vector space over $\mathbb{C}$.

$\mathcal{R} = C^\infty(I, \mathbb{C})$, where $I \subseteq \mathbb{R}$ is open interval.

$D: C^\infty \rightarrow C^\infty: f \mapsto f'$, $\mathbb{C}$ -linear map, by linearity of differential,

$M_a: C^\infty \rightarrow C^\infty: f \mapsto af$, $\mathbb{C}$ -linear map, C^∞ -module Homomorphism.

$$\begin{aligned} L(y) &= a_n(t) \cdot y^{(n)} + a_{n-1}(t) y^{(n-1)} + \dots + a_1(t) y' + a_0(t) y \\ &= (M_0 D^n + M_1 D^{n-1} + \dots + M_{n-1} D + M_n) y. \end{aligned}$$

Prop. $D: \mathcal{R} \rightarrow \mathcal{R}; f \mapsto f'$ is linear operator.

Proof. Let $f, g \in \mathcal{R} = C^\infty(I, \mathbb{C})$ and $c \in \mathbb{C}$ be given. Then, for any $x \in I$,

$$\begin{aligned} D(cf+g)(x) &= \lim_{h \rightarrow 0} \frac{c \cdot f(x+h) + g(x+h) - (c \cdot f(x) + g(x))}{h} \\ &= c \cdot \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\ &= c \cdot D(f)(x) + D(g)(x) \quad \square \end{aligned}$$

Prop. $M_\alpha: \mathcal{R} \rightarrow \mathcal{R}; f \mapsto \alpha f$ is linear operator for any $\alpha(t) \in \mathcal{R}$.

Proof. Let $f, g \in \mathcal{R} = C^\infty(I, \mathbb{C})$ and $c \in \mathbb{C}$ be given. Then, for any $x \in I$,

$$M_\alpha(cf+g)(x) = \alpha(x) \cdot c \cdot f(x) + \alpha(x) \cdot g(x) = c \cdot M_\alpha(f)(x) + M_\alpha(g)(x). \quad \square$$

Prop. For any $\alpha \in \mathcal{R}$,

$$DM_\alpha = M_\alpha D + M_{D\alpha} \quad (\text{That is, } [D, M_\alpha] = M_{D\alpha}).$$

Proof. Given $f \in \mathcal{R}$,

$$DM_\alpha(f) = D(\alpha f) = \alpha' f + \alpha f' = M_{\alpha'}(f) + M_\alpha D(f).$$

$$\Gamma_{hm}. \text{ On } B = C^\infty(I, \mathbb{C}),$$